

Parthiv Gajula

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PROFESSIONAL SUMMARY

Focused on **large language models**, with hands-on experience training and modifying models from **scratch** while studying their **architectures** and behavior in real systems. Built and deployed end-to-end applications in an **Agile** development environment, with a focus on applying LLMs to real-world problems.

EDUCATION

Wayne State University — Detroit, MI

Aug 2023 – May 2026

B.S. Computer Science, Minor in Artificial Intelligence

Relevant Coursework: Introduction to Deep Learning, Introduction to Large Language Models, Intelligent Systems and Data Mining, Operating Systems

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, C++, Kotlin, Ruby, R

Frameworks: React, Next.js, Flask, Ruby on Rails, PyTorch

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase

DevOps/Cloud: Docker, Kubernetes, Terraform, Jenkins, Nginx, Render

Tools: Postman, SSMS, Expo

Testing: Jest, Selenium, E2E, ESLint

Security: OAuth 2.0, JWT, OpenSSL, AES, Apache Hardening

AI/ML: Retrieval-Augmented Generation(RAG), Transformers, BERT, GPT, Fine-Tuning, Prompt Engineering, Encoder-Decoder Models: LSTM, RNNs, CNNs, GANs, VAEs, RLHF, Multimodal Models

WORK EXPERIENCE

365 Retail Markets — Database Intern

May 2025 – Aug 2025

- Optimized SQL queries, improving execution speed by up to 45% through indexing and execution plan analysis
- Managed remote database servers and configured replication with filtering across instances to ensure data consistency
- Improved query performance by analyzing ad-hoc queries, prepared statements, and execution plans
- Developed SQL triggers to automate data processing and enforce data integrity

Tech: MySQL, SQL Server, SSMS, Replication, Query Optimization

PROJECTS

SpeechMatch – Capstone Team Lead

Jan 2026 – Present

- Led a capstone team developing a mobile pronunciation training platform that evaluates spoken audio against reference speech
- Collaborated directly with the project client to gather requirements, refine features, and align development milestones
- Designed backend APIs for audio processing, OCR-based passage ingestion, and pronunciation evaluation

Tech: FastAPI, React Native, PostgreSQL, OCR, MFA, Azure Voice Services, Docker

AWS S3 Encrypted FUSE Filesystem

Sep 2025 – Dec 2025

- Built a FUSE3-based filesystem that encrypts files before upload to an S3-compatible bucket
- Intercepted the FUSE file descriptor cache (fdcache) to insert write I/O operations prior to upload
- Secured data at rest using AES-256 encryption via OpenSSL
- Designed a SHA-256 hashing pipeline to ensure data integrity and backward compatibility
- Developed a remote directory abstraction layer for testing encrypted uploads

Tech: AWS S3, C, FUSE3, OpenSSL, Linux, AES-256

Spotify Background Recommendation App

Oct 2025 – Jan 2026

- Built an ML-based recommendation system that analyzes listening history to generate personalized playlists
- Applied clustering and similarity metrics to model evolving user preferences
- Integrated Spotify Web API for real-time track data and playback analytics

Tech: Python, Spotify Web API, Flask, K-Means Clustering, OAuth 2.0

DriveMate - AI Driving Assistant | UM-Dearborn Hackathon (Winner: Best Use of ElevenLabs)

Oct 2025

- Led development of DriveMate, a real-time conversational AI co-pilot designed to keep drivers engaged and alert
- Built a low-latency voice pipeline integrating speech-to-text, generative AI (Fetch.ai ASI-1), and text-to-speech for real-time interaction
- Implemented a multi-agent architecture using Python uAgents for context-aware routing, memory, and response generation

Tech: Flask, Web Speech API, Eleven Labs TTS, Fetch.ai ASI-1, uAgents

Retrieval Augmented Generation (RAG) Research Study | parthivg.com/rag

Jan 2025 – May 2025

- Explored **prompt design** and retrieval strategies to improve generation quality
- Implemented **RAG pipelines** using DPR, ColBERT, and SELF-RAG architectures
- Built **end-to-end** retrieval systems with document chunking, indexing, and embedding-based search
- Benchmarked **RAG vs LLM**-only generation using MS MARCO, analyzing accuracy and retrieval effectiveness
- Investigated **ranking behavior** and **failure modes** in dense retrieval systems, including hallucination and context relevance limitations
- Explored **prompt design** and retrieval strategies to improve generation quality and grounding

Tech: DPR, ColBERT, SELF-RAG, MS MARCO, Sequence to Sequence models

Ubuntu System Administration Project

Sep 2024 – Dec 2024

- Configured an Ubuntu server with SSL/TLS over HTTPS on a non-standard port (444)
- Generated and deployed custom SSL certificates using OpenSSL
- Applied security hardening to Apache2 services

Tech: Ubuntu Linux, Apache2, OpenSSL, OpenVAS, Kali Linux, Bash

HarmonAlze - Music Recommendation System | Oakland University Hackathon (Winner: Interactive Media)

Feb 2024

- Developed a full-stack music recommendation system using heart-rate (BPM) data
- Improved BPM prediction accuracy by **80% using K-Means clustering**
- Processed audio features with librosa and integrated Fitbit biometric data

Tech: PyTorch, Librosa, Fitbit API, Flask, K-Means Clustering